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AGRICULTURE

Geography - Class 10 | Chapter 4 | Deep
Structured NCERT Notes

1. Introduction to Agriculture

What is Agriculture?

Agriculture is a primary activity which means it is **directly dependent on nature**. It involves **cultivation of crops, rearing of animals, fishing, and forestry**. Agriculture is the **main source of livelihood** for about **58% of India's population**.

Agriculture is the backbone of India's economy as it contributes **about 17% to India's Gross Domestic Product (GDP)**. Around **58% of India's population depends on agriculture** for their livelihood. The **performance of agriculture affects the overall economic growth** of the country.

Agriculture provides food for the population, raw materials for industries, and employment to millions of people. It is essential for food security of the nation. Without agriculture, human civilization cannot survive.

Factors Affecting Agriculture

Agriculture depends on several physical and human factors:

Physical Factors:

- 1. Climate:** Climate is the most important factor affecting agriculture. **Temperature, rainfall, humidity, and sunlight** determine **what crops can be grown** in a region. Different crops require **different climatic conditions** for their growth.
- 2. Soil:** Soil is the medium for plant growth. Different crops require **different types of soil**. **Alluvial soil, black soil, red soil, and laterite soil** are suitable for **different crops**. Soil **fertility affects crop yield**.
- 3. Topography:** The shape and slope of land affects agriculture. **Flat plains are suitable for cultivation** while **steep slopes are not**. **Terrace farming** is practiced on **hilly slopes** to cultivate crops.

Human Factors:

1. Population: Population provides labor for agricultural activities. **Dense population** means **more labor availability** but also **more pressure on land**. **Labor-intensive farming** is practiced in **densely populated areas**.

2. Technology: Modern technology improves agricultural productivity. **High-yielding variety (HYV) seeds, chemical fertilizers, pesticides, and machinery** increase **crop yields**. **Technology** reduced the **drudgery** of agricultural work.

3. Market: **Access to markets** determines **what crops farmers grow**. Farmers grow crops that **fetch good prices in the market**. **Transportation facilities** help farmers **sell their produce** easily.

4. Government Support: **Government policies** affect agriculture. **Minimum support price (MSP), subsidies, crop insurance, and agricultural loans** support farmers. **Government research institutes** develop **new crop varieties**.

2. Types of Farming

Primitive Subsistence Farming

Primitive subsistence farming is practiced on small patches of land using primitive tools like hoe, dao, and digging sticks. This type of farming depends on monsoon rainfall and natural soil fertility. Family labor is used for cultivation.

This farming is also called 'slash and burn' agriculture. Farmers clear a patch of land by cutting and burning vegetation, then cultivate crops on the ash-enriched soil. When soil fertility decreases, farmers move to a new patch of land.

Primitive subsistence farming is practiced in scattered areas with low population density. It is practiced in parts of Northeast India (Jhumming), Central India, and Southern India. Yields are low as no modern inputs are used.

Key characteristics of primitive subsistence farming:

- **Small land holdings** (less than 1 hectare)
- **Primitive tools** (hoe, dao, digging sticks)
- **Depends on monsoon and natural soil fertility**

- **Family labor** only
- **Low productivity**
- **Crops grown for self-consumption** (subsistence)
- **No use of modern inputs** (fertilizers, pesticides, HYV seeds)

Primitive Subsistence Farming - Regional Names

Different regions in India have different names for slash and burn agriculture:

Jhumming is practiced in **Northeast India** (Assam, Meghalaya, Mizoram, Nagaland). It involves **cutting and burning forest vegetation**, then **cultivating crops on the ash**. After **2-3 years**, soil fertility decreases and farmers **move to new land**.

Pamlou is practiced in **Manipur**. It is similar to jhumming where **forest land is cleared by burning** and crops are cultivated.

Dakon is practiced in **Meghalaya**. Farmers **clear forest patches** and cultivate crops using **traditional methods**.

Bewar or **Dahiya** is practiced in **Madhya Pradesh**. Forest land is **cleared by burning** and crops are grown.

Podu or **Penda** is practiced in **Andhra Pradesh and Odisha**. Farmers **clear forest patches** and practice **slash and burn agriculture**.

Valre or **Waltre** is practiced in **south-east Rajasthan**. It involves **clearing forest land** for cultivation.

Kumari is practiced in **Western Ghats**. Forest patches are **cleared and burned** for cultivation.

Khil is practiced in **Himalayan belt**. Similar slash and burn method is used.

Kharif crops are often grown in these systems after clearing land.

Intensive Subsistence Farming

Intensive subsistence farming is practiced in areas with high population density on small land holdings. Farmers use **more labor and capital to maximize output from small land areas**. This farming is **labor-intensive** rather than **capital-intensive**.

In intensive subsistence farming, farmers use modern inputs like HYV seeds, chemical fertilizers, pesticides, and irrigation to increase productivity. Multiple cropping is practiced to maximize land use.

This type of farming is practiced in densely populated areas of Indian plains like Uttar Pradesh, Bihar, Punjab, Haryana, and West Bengal. Land holdings are small due to population pressure.

Key characteristics of intensive subsistence farming:

- **High population density**
- **Small land holdings**
- **Labor-intensive farming**
- **Use of modern inputs**
- **Multiple cropping**
- **High productivity per unit area**
- **Crops grown mainly for family consumption with small surplus for sale**

Commercial Farming

Commercial farming is practiced where crops are grown and animals are reared for sale in the market. The main objective is profit rather than self-consumption. Large capital and modern inputs are used to maximize production.

In commercial farming, HYV seeds, chemical fertilizers, pesticides, and machinery are used

extensively. **Irrigation is reliable** and **large-scale production** is achieved. **Farmers grow crops based on market demand and prices.**

Commercial farming is practiced in areas with good infrastructure like roads, railways, and markets.

States like **Punjab, Haryana, and western Uttar Pradesh** practice **commercial farming** extensively. **Plantation agriculture is a type of commercial farming.**

Key characteristics of commercial farming:

- **Crops grown for sale** in the market
- **Large land holdings**
- **High use of modern inputs**
- **Reliable irrigation**
- **High productivity**
- **Profit-oriented**
- **Good market access**

Plantation Agriculture

Plantation agriculture is a type of commercial farming where a single crop is grown on a large estate. Plantations are large farms that cover hundreds or thousands of hectares. It is capital-intensive with large investments in land, machinery, and labor.

In plantation agriculture, crops like tea, coffee, rubber, sugarcane, banana, and cotton are grown. These crops require fertile soil, adequate rainfall, and warm climate. Plantations are usually located in tropical and subtropical regions.

Plantation agriculture requires significant capital investment for land development, planting material, irrigation, and processing facilities. Labor is employed for planting, maintenance, and harvesting. Products are processed on the plantation or in nearby factories.

Key characteristics of plantation agriculture:

- **Single crop grown on large estate**
- **Large land holdings**
- **Capital-intensive**

- **Requires fertile soil, adequate rainfall, warm climate**
- **Labor employed for cultivation and harvesting**
- **Products processed on plantation or nearby factories**
- **Crops grown for export and domestic market**
- **Good transportation facilities needed**

Important plantation crops in India:

- **Tea:** Assam, West Bengal (Darjeeling), Tamil Nadu, Kerala
- **Coffee:** Karnataka, Kerala, Tamil Nadu
- **Rubber:** Kerala, Tamil Nadu, Karnataka
- **Sugarcane:** Uttar Pradesh, Maharashtra, Karnataka
- **Banana:** Tamil Nadu, Maharashtra, Gujarat

3. Cropping Seasons in India

Rabi Crops

Rabi crops are sown in winter from October to December and harvested in summer from April to June. These crops **require cool growing season and bright sunshine at the time of ripening.** Rabi crops **depend on irrigation as winter rainfall is limited.**

Major Rabi crops include wheat, barley, peas, gram, and mustard. Wheat is the most important Rabi crop in India. India is the **second largest producer of wheat** in the world after China.

Wheat requires cool and moist climate during growing season. It needs **50-75 cm of rainfall** and **well-distributed rainfall.** **Alluvial soil and black soil** are suitable for wheat cultivation. Major wheat-producing states are **Punjab, Haryana, Uttar Pradesh, Bihar, Rajasthan, and Madhya Pradesh.**

Gram, peas, and mustard are also important Rabi crops. They require **moderate rainfall and temperature** and are grown widely in **Madhya Pradesh, Uttar Pradesh, Maharashtra, Rajasthan, Punjab, Haryana, and Gujarat.**

Key characteristics of Rabi crops:

- **Sown: October to December**
- **Harvested: April to June**
- **Require cool growing season**
- **Bright sunshine at ripening time**
- **Depend on irrigation**
- **Major crops:** Wheat, barley, peas, gram, mustard

Kharif Crops

Kharif crops are sown with the onset of monsoon from June to July and harvested in September to October. These crops **require warm and wet climate** for their growth. **Kharif crops depend mainly on monsoon rainfall.**

Major Kharif crops include rice, maize, cotton, sugarcane, bajra, jowar, tur, moong, urad, and groundnut. Rice is the most important Kharif crop in India. India is the second largest producer of rice in the world after China.

Rice requires high temperature above 25°C and high humidity. It needs **100-200 cm of rainfall.** **Alluvial soil in plains and deltaic regions** is suitable for rice cultivation.

Cotton requires high temperature and light rainfall. It needs **210 frost-free days** and **50-100 cm rainfall**. **Black soil** is ideal for cotton cultivation.

Sugarcane requires hot and humid climate. It needs **75-100 cm rainfall** and can grow on **various types of soil**. Major sugarcane-producing states are **Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, and Andhra Pradesh**.

Key characteristics of Kharif crops:

- **Sown: June to July**
- **Harvested: September to October**
- **Require warm and wet climate**
- **Depend mainly on monsoon rainfall**
- **Major crops:** Rice, maize, cotton, sugarcane, bajra, jowar, tur, moong, urad, groundnut

Zaid Season

Zaid is a short season between Rabi and Kharif from **March to June**. This season is **shorter than Rabi and Kharif seasons**. Zaid crops are **grown during summer months**.

Zaid crops include watermelon, muskmelon, cucumber, vegetables, and fodder crops. These crops

require warm and dry weather with **shorter day length**. **Irrigation is essential** as there is **no monsoon rainfall** during this season.

Zaid season allows farmers to grow additional crops and increase their income. This season is **important for vegetable production**. States like **Uttar Pradesh, Bihar, and West Bengal** practice Zaid farming extensively.

Key characteristics of Zaid season:

- **Duration: March to June**
- **Short season**
- **Require warm and dry weather**
- **Irrigation essential**
- **Major crops:** Watermelon, muskmelon, cucumber, vegetables, fodder

4. Major Crops of India

Food Crops

Food crops are grown for human consumption. They include **cereals, pulses, and fruits/vegetables**. Food crops are **essential for food security** of the nation.

Rice is the most important food crop of India. India is the **second largest producer of rice** in the world after China. Rice provides a major share of food grain production in India.

Rice requires high temperature above **25°C** and **high humidity**. It needs **100-200 cm of rainfall**. **Alluvial soil in plains and deltaic regions** is suitable for rice cultivation.

Major rice-producing states:

- **West Bengal**
- **Uttar Pradesh**
- **Punjab**
- **Haryana**
- **Andhra Pradesh**
- **Tamil Nadu**
- **Odisha**
- **Bihar**

Wheat is the second most important food crop of India. India is the **second largest producer of wheat** in the world after China. Wheat requires **cool and moist climate** during growth and **bright sunshine at ripening time**.

Major wheat-producing states:

- **Uttar Pradesh**
- **Punjab**
- **Haryana**
- **Madhya Pradesh**
- **Bihar**
- **Rajasthan**

Maize, bajra, and jowar are also important food crops. They are used for **food, fodder, and industrial purposes** and are grown in different climatic regions.

Pulses

Pulses are important source of protein in vegetarian diet. They are **rich in protein** and **essential for nutrition**. India is the **largest producer and consumer of pulses** in the world.

Pulses help in maintaining soil fertility as they **fix nitrogen from atmosphere**. This **reduces the need for chemical fertilizers**. Pulses are **rotated with other crops to improve soil health**.

Major pulses grown in India:

- **Gram:** important Rabi pulse grown in Madhya Pradesh, Uttar Pradesh, Maharashtra, and Rajasthan.
- **Tur:** important pulse grown in Maharashtra, Karnataka, Andhra Pradesh, and Madhya Pradesh.
- **Moong:** short-duration pulse grown in Rajasthan, Uttar Pradesh, Maharashtra, and Karnataka.
- **Urad:** warm and humid climate pulse grown in Andhra Pradesh, Tamil Nadu, Karnataka, and Maharashtra.
- **Masoor:** Rabi pulse grown in Uttar Pradesh, Madhya Pradesh, Rajasthan, and Bihar.

Oilseeds

Oilseeds are grown for extracting oil for cooking, industrial uses, and soap manufacturing. Oilseeds are **important cash crops** for farmers. India is the **second largest producer of oilseeds** in the world.

Groundnut is the most important oilseed crop. It contains **about 50% oil**. Groundnut requires **moderate temperature and rainfall** and **sandy loam soil**.

Major groundnut-producing states:

- **Gujarat**
- **Andhra Pradesh**

- **Tamil Nadu**
- **Karnataka**
- **Maharashtra**

Mustard, sesamum, sunflower, and soybean are also important oilseeds. They are grown for **edible oil** and **industrial use**.

Fiber Crops

Fiber crops are grown for their fibers used in **textile industry**. Fiber crops are **important cash crops** for farmers.

Cotton is the most important fiber crop. India is the **origin of cotton** and the **third largest producer** in the world. Cotton requires **high temperature, light rainfall, 210 frost-free days, and 50-100 cm rainfall**.

Black soil is ideal for cotton cultivation. Cotton is a **Kharif crop** taking **6-8 months to mature**.

Major cotton-producing states:

- **Maharashtra**
- **Gujarat**
- **Madhya Pradesh**
- **Karnataka**

- **Andhra Pradesh**
- **Tamil Nadu**
- **Punjab**
- **Haryana**

Jute is known as the 'golden fiber'. India is the **largest producer of jute** in the world. Jute requires **high temperature, heavy rainfall, humid climate**, and **alluvial soil**.

Major jute-producing states:

- **West Bengal**
- **Bihar**
- **Assam**
- **Odisha**
- **Uttar Pradesh**

Plantation Crops

Plantation crops are grown on large estates as commercial crops. These crops require **significant capital investment and labor**. Plantation crops are **important for export and domestic market**.

Tea is an important plantation crop. India is the **second largest producer of tea** in the world after China.

Tea requires **warm and moist climate, 150-300 cm rainfall**, and **deep, friable, well-drained soil**.

Major tea-producing states:

- **Assam**
- **West Bengal**
- **Tamil Nadu**
- **Kerala**
- **Himachal Pradesh**
- **Uttarakhand**

Coffee is an important plantation crop. India produces **Arabica and Robusta varieties**. Coffee requires **warm and wet climate, 150-250 cm rainfall**, and **well-drained volcanic soil**.

Major coffee-producing states:

- **Karnataka**
- **Kerala**
- **Tamil Nadu**

5. Agricultural Reforms in India

Green Revolution

Green Revolution refers to the significant increase in food grain production due to adoption of modern agricultural technology. It started in **1960s** in India. Green Revolution made India **self-sufficient in food grains**.

Green Revolution involved use of HYV seeds, chemical fertilizers, pesticides, and irrigation. Modern machinery like tractors, harvesters, and threshers were introduced. Chemical inputs increased productivity significantly.

Green Revolution was successful in wheat and rice cultivation. Punjab and Haryana became the breadbasket of India. Food grain production increased sharply after the Green Revolution.

Benefits of Green Revolution:

- **Increased food grain production**
- **India became self-sufficient in food**
- **Increased farmer income** in Green Revolution areas
- **Reduced hunger and famine**

- **Created employment** in agriculture and related sectors

Problems of Green Revolution:

- **Regional imbalance**
- **Soil degradation**
- **Groundwater depletion**
- **Increased cost of cultivation**
- **Small farmers could not afford modern inputs**
- **Environmental pollution**

White Revolution (Operation Flood)

White Revolution refers to the significant increase in milk production in India. It was launched by **Dr. Verghese Kurien**, who is known as the **father of White Revolution**. **Operation Flood** was the **world's largest agricultural program**.

White Revolution made India the largest milk producer in the world. **Amul cooperative model** was successful in **Gujarat**. **National Dairy Development Board (NDDB)** promoted dairy development.

White Revolution benefited millions of small farmers by providing **market for their milk**. Dairy farming became **important source of income** for rural

families. India produces **about 20% of world's milk**.

Other Agricultural Reforms

Land Reforms: **Land ceiling acts** limited the **maximum land a person could own**. **Land consolidation** combined **small fragmented holdings** into **larger farms**. **Abolition of intermediaries** gave **land rights to tillers**.

Minimum Support Price (MSP): **Government announces MSP** for major crops before sowing season. **MSP protects farmers from price fall**. MSP ensures **remunerative prices** for farmers.

Kisan Credit Card (KCC): **KCC provides affordable credit** to farmers. It helps farmers **purchase seeds, fertilizers, and other inputs**. KCC has **reduced dependence on moneylenders**.

Crop Insurance: **Crop insurance protects farmers** from **crop failure due to natural calamities**. **Pradhan Mantri Fasal Bima Yojana (PMFBY)** provides **affordable crop insurance**. Insurance helps farmers **recover from losses**.

Agricultural Markets: **APMC mandis** provide **market for agricultural produce**. **e-NAM** is an **online trading platform**. E-NAM helps farmers **get better prices**.

6. Contribution of Agriculture to Economy

Agriculture's Role in Indian Economy

Agriculture contributes about 17% to India's GDP. Despite this **relatively low contribution**, **58% of India's population depends on agriculture** for livelihood. This shows **low productivity per worker** in agriculture.

Agriculture provides raw materials to industries. **Textile industry** depends on **cotton and jute**. **Sugar industry** depends on **sugarcane**. **Food processing industry** depends on **various agricultural products**.

Agriculture provides employment to rural population. **About 58% of workforce** is employed in agriculture. **Agriculture is main source of income** for **rural families**.

Agriculture contributes to food security of the nation. India is **self-sufficient in food grains**. **Food grain production** has increased massively since independence.

Agriculture contributes to exports. **Rice, wheat, cotton, sugar, and spices** are **major agricultural exports**. Agricultural exports earn **foreign exchange** for the country.

Challenges Facing Indian Agriculture

- **1. Small and Fragmented Land Holdings:** Average land holding is small and fragmented due to inheritance laws.
- **2. Dependence on Monsoon:** A large part of Indian agriculture is **rain-fed** and depends on monsoon rainfall.
- **3. Low Productivity:** Yield per hectare is low compared to international standards.
- **4. Lack of Storage and Cold Storage:** Post-harvest losses are high because storage facilities are inadequate.
- **5. Poor Market Access:** Farmers cannot always get fair prices for their produce.
- **6. Indebtedness:** Farmers borrow at high interest rates because cultivation costs are high.
- **7. Climate Change:** Changing rainfall patterns, increased temperature, droughts, floods, and unseasonal rainfall affect crops.
- **8. Soil Degradation:** Excessive fertilizers, erosion, salinity, and unsustainable farming reduce soil fertility.

7. Agricultural Technology and Modernization

Modern Agricultural Technology

Modern agricultural technology includes HYV seeds, chemical fertilizers, pesticides, and machinery. HYV seeds produce higher yields than traditional seeds. **Chemical fertilizers provide nutrients** to plants quickly.

Tractors, harvesters, threshers, and irrigation pumps are modern agricultural machinery. Machinery reduces drudgery and increases efficiency. Mechanization is increasing in Indian agriculture.

Irrigation technology includes tube wells, drip irrigation, and sprinkler irrigation. Tube wells provide reliable water source. Drip and sprinkler irrigation save water.

Organic Farming

Organic farming avoids use of chemical fertilizers, pesticides, and HYV seeds. It uses organic manure, natural pesticides, and traditional seeds. Organic farming is environmentally friendly and sustainable.

Organic farming produces healthier food without **chemical residues**. **Organic products fetch higher prices** in the market. **Consumer demand for organic food is increasing**.

India has large potential for organic farming. **Sikkim became India's first organic state**. **Government promotes organic farming** through **Paramparagat Krishi Vikas Yojana (PKVY)**.

Precision Farming

Precision farming uses technology to optimize crop production. It uses **GPS, sensors, drones, and data analytics** to **monitor and manage crops**. Precision farming **maximizes productivity while minimizing inputs**.

Precision farming helps in precise application of water, fertilizers, and pesticides. This **reduces waste and cost**. **Water and fertilizer use efficiency increases**.

Precision farming is still in early stages in India. It is mainly practiced in **commercial farming** and **plantation crops**. **Technology cost is high** for small farmers.

8. Agriculture and Globalization

Impact of Globalization on Indian Agriculture

Globalization has both positive and negative impacts on Indian agriculture. Positive impacts include access to international markets, import of modern technology, and improved farming practices. Negative impacts include competition from imported agricultural products, price volatility, and threat to small farmers.

Globalization has increased export opportunities for Indian agricultural products. Rice, wheat, cotton, sugar, and spices are major exports. Indian agriculture is integrated with global markets.

Globalization has exposed Indian farmers to international competition. Subsidized agricultural products from developed countries compete with Indian products. Small farmers face difficulties competing with large-scale foreign producers.

WTO agreements affect Indian agriculture. Agreement on Agriculture (AoA) regulates agricultural trade. India needs to balance farmer interests with trade commitments.

9. Key Agricultural Statistics

Important Facts About Indian Agriculture

Agriculture contributes about 17% to India's GDP. 58% of India's population depends on agriculture for livelihood. India has 2.4% of world's land area but supports 18% of world's population.

Food grain production:

- **1950-51:** 50 million tonnes
- **2020-21:** Over 250 million tonnes
- **Increase:** 5 times in 70 years

India's position in world production:

- **Rice:** 2nd largest producer
- **Wheat:** 2nd largest producer
- **Pulses:** Largest producer
- **Oilseeds:** 2nd largest producer
- **Cotton:** 3rd largest producer
- **Jute:** Largest producer
- **Tea:** 2nd largest producer
- **Milk:** Largest producer

Land use pattern:

- **Net sown area:** 43% of total geographical area
- **Forest:** 23% of total geographical area
- **Barren and wasteland:** 5% of total geographical area
- **Culturable wasteland:** 5% of total geographical area
- **Current fallow:** 8% of total geographical area
- **Other fallow:** 4% of total geographical area

Irrigation sources:

- **Canals:** 24% of irrigated area
- **Wells and tube wells:** 63% of irrigated area
- **Other sources:** 13% of irrigated area

Land holding size:

- **Average land holding:** 1.08 hectares
- **Small and marginal farmers:** 86% of total farmers
- **Semi-medium and medium farmers:** 13% of total farmers
- **Large farmers:** 1% of total farmers

10. Summary of Chapter 4: Agriculture

Key Points from NCERT

- **1. Agriculture is a primary activity:** Directly dependent on nature. 58% of population depends on agriculture. Contributes 17% to GDP.
- **2. Types of farming:** Primitive subsistence, intensive subsistence, commercial farming, and plantation agriculture.
- **3. Cropping seasons:** Rabi, Kharif, and Zaid are the three major cropping seasons.
- **4. Major crops:** Food crops, pulses, oilseeds, fiber crops, and plantation crops.
- **5. Agricultural reforms:** Green Revolution, White Revolution, land reforms, MSP, KCC, crop insurance, and e-NAM.
- **6. Challenges:** Small land holdings, monsoon dependence, low productivity, storage issues, market access, indebtedness, climate change, and soil degradation.
- **7. India's agricultural position:** Major producer of rice, wheat, pulses, milk, jute, and tea.

11. Important Terms from NCERT

Important Terms

- **Primary Activity:** Activity directly dependent on nature, such as agriculture, fishing, and forestry.
- **Subsistence Farming:** Farming for self-consumption, not mainly for sale.
- **Commercial Farming:** Farming for sale in market and profit.
- **Slash and Burn:** Clearing land by cutting and burning vegetation.
- **HYV Seeds:** High-Yielding Variety seeds that produce higher yields.
- **Green Revolution:** Significant increase in food grain production through modern technology.
- **White Revolution:** Significant increase in milk production through Operation Flood.
- **Multiple Cropping:** Growing 2-3 crops per year on the same land.
- **Rabi Crops:** Sown from October to December and harvested from April to June.
- **Kharif Crops:** Sown from June to July and harvested from September to October.

- **Zaid Season:** Short season between Rabi and Kharif, from March to June.
- **MSP:** Minimum Support Price, a government-guaranteed price for crops.
- **KCC:** Kisan Credit Card, which provides affordable credit to farmers.
- **Plantation Agriculture:** Single crop grown on a large estate with capital investment.